# MURAD HOSSAIN

Gazipur, Bangladesh | +880 1682675115 | hossainmurad.me@gmail.com
LinkedIn: Murad Hossain | ResearchGate: Murad-Hossain-26

## RESEARCH INTERESTS

- Refrigeration, vacuum technology and thermal testing
- Process development and automation in manufacturing
- Simulation and topology optimization
- IoT irrigation, climate-smart agriculture and low-carbon waste management

## EDUCATION

**Bachelor of Science in Mechanical Engineering** *Jan 2015 to Oct 2019*

*Rajshahi University of Engineering & Technology (RUET), Rajshahi, Bangladesh*

Thesis: Simulation of Modified Flipped Airfoil and Analysis of Lift and Drag Coefficient. Supervisor: Dr. Md. Emdadul Hoque.

Coursework: Mechanical Engineering Drawing, Engineering Mechanics, Computer Programming, Mechanics of Solids, Heat Transfer, Fluid Mechanics, Engineering Metallurgy, Automobile Engineering, Refrigeration & Air Conditioning.

## RESEARCH & PUBLICATIONS

**FEA-Driven Topology Optimization for Lightweight Structural Design: A Computational Framework with Implicit Geometry Reconstruction** *2026*

Afsar Mahmud, Hridoy Bhadra, Murad Hossain. Topology optimized Zamak 3 refrigerator hinge bracket with 45.7% less mass, checked with FEA and rebuilt for manufacturing (DfM). International Conference on Mechanical, Industrial and Materials Engineering.

**A Greener Future: IoT-Powered Smart Irrigation for Bangladesh** *2025*

Md. Foysal, Mehedi Hasan, Murad Hossain, Most Salma Khatun. An IoT based automated irrigation system that uses water better and raises crop yield where water is scarce and rainfall is unpredictable. 1st International Conference on Agricultural Machinery and Bioresource Engineering (ICAMBE) 2025, Bangladesh Agriculture University.

**Dynamics of Municipal Solid Waste in Urban Bangladesh: Compositional Shifts, GHG Emissions, and Policy Implications (2005-2041)** *2025*

Md. Foysal, Most Salma Khatun, Mehedi Hasan, Murad Hossain, Zannatul Fhardous Umama, Md. Fahad Hossain. Estimates a 324% rise in waste generation by 2041 at current GDP growth, with Chittagong and Dhaka as the main hotspots. The results give policymakers a scientific basis for low-carbon waste management. 4th National Research Project Contest 2025, Bangladesh University of Professionals.

**Leveraging Climate-Smart Agriculture for Mitigation: Sustainable Solutions from Bangladesh** *2025*

Md. Foysal, Most Salma Khatun, Mehedi Hasan, Murad Hossain, Zannatul Fhardous Umama. Looks at climate-smart agriculture, which combines adaptation and mitigation so farmers can raise productivity and cope with shifting rainfall. IAAS Asia Pacific Scientific Event 2025.

**Simulation of Modified Flipped Airfoil and Analysis of Lift and Drag Coefficient** *2019*

Murad Hossain, Prashanta Saha. Simulation of a NACA BE50 airfoil with different-sized dimple surfaces at different orientations, tested at different angles of attack to find the orientation with the least drag. Thesis paper, Rajshahi University of Engineering & Technology.

## PROFESSIONAL EXPERIENCE

**Senior Assistant Director, Process Development (Refrigerator Division)** *May 2021 to Present*

*Walton Hi-Tech Industries PLC., Gazipur, Bangladesh. Promoted from Deputy Assistant Director to Senior Assistant Director.*

- Designed and improved cooling-performance testing, vacuum control to reduce choking, refrigerant charging and selection, tube joining, traceability and material-handling procedures for refrigerator manufacturing.
- Designed a cooling-circuit temperature based decision system that cuts aging time and saves 158.4 MW of annual power consumption.
- Set up an aging test model based on thermal imaging that saves 117.5 MW of electricity per year.
- Installed an IR based industrial thermal-gun temperature measurement model. It cut testing time by 15 minutes and saves 69.71 MW of energy per year.

- Installed a Pirani-gauge monitoring system with CAN-HS that gives real-time vacuum data. Choking problems fell from about 0.13% in 2021 to 0.02% in 2025.
- Installed an oil-pulse pneumatic gun for better operator ergonomics. Loosening torque rose from 7 Nm to 10 Nm, a 43% improvement.
- Designed a thinner PET belt with 43% higher thermal joint strength and no extra material.
- Ran a 60-foot AGV pilot project, including payload analysis and lightweight box design.

**Executive Engineer, Production Planning & Design (Boiler)** *Nov 2019 to Apr 2021*

*New Model Boiler Company Limited, Gazipur, Bangladesh. Promoted from Provisionary Engineer to Executive Engineer.*

- Designed shell type and fire-tube boilers up to 1,000 kg capacity following BBR (1951) and ASME standards.
- Supervised fabrication, radiographic and ultrasonic testing, heat treatment, and hydraulic testing of boilers.
- Designed and commissioned a steam piping network that carries steam to industrial vapor-iron units at garment factories, sizing and routing the pipeline to cover the full length of the finishing section.
- Trained marketing officers on boiler engineering.

**Internship, Bangladesh Industrial Technical Assistance Centre (BITAC)** *May 2018 to Jun 2018*

*Dhaka, Bangladesh*

- Hands-on training in the Machine Tools Shop, CNC Shop, Heat Treatment, Foundry, Welding & Fabrication, and the Tool & Technology Institute (TTI).

## TECHNICAL SKILLS

- Focus area: Refrigeration and vacuum technology, thermal imaging, process automation, FEA
- CAD and simulation: SolidWorks, AutoCAD, Ansys
- Programming: Python, C, HTML and CSS
- Quality and process: Six Sigma Yellow Belt, DFM/DFA, traceability
- Software: Microsoft Office

## MAJOR PROJECTS

**Cooling Circuits Temperature Profile Based Decision Making** *Jul 2025 to Present*

Cuts aging time and saves 158.4 MW of annual power consumption.

**Thermal Image Based Aging Test Model** *Aug 2024 to May 2025*

Saves 117.5 MW of electricity per year.

**Pirani Gauge with CAN-HS System** *Jan 2024 to Jun 2025*

Real-time vacuum data cut choking from about 0.13% (2021) to 0.02% (2025).

**60-Foot AGV Pilot Project** *Aug 2023 to Feb 2024*

Payload analysis and lightweight box design.

**IR Based Industrial Thermal Gun Temperature Measurement** *Oct 2021 to Jul 2023*

Cut testing time by 15 minutes and saves 69.71 MW of energy per year.

**Oil Pulse Pneumatic Gun Implementation** *Aug 2025 to Dec 2025*

Loosening torque up from 7 Nm to 10 Nm (43%), with better operator ergonomics.

**PET Belt Joining Strength Improvement** *Jul 2021 to Dec 2021*

Thinner PET belt with 43% higher thermal joint strength and no extra material.

**Steam Piping Network for Vapor-Iron Units**

Sized, routed and commissioned the piping that carries steam across the finishing section of a garment factory.

**Simulation of Modified Flipped Airfoil** *Aug 2018 to Aug 2019*

NACA BE50 with different-sized dimple surfaces at different orientations, analyzed at different angles of attack to find the best orientation for lower drag.

## CERTIFICATIONS & TRAINING

- Six Sigma Yellow Belt, 6Sigma Study (Nov 2024)
- SOLIDWORKS Design Associate (CSWA), Dassault Systèmes (Nov 2024)
- SOLIDWORKS Additive Manufacturing Associate, Dassault Systèmes (Nov 2024)

- SOLIDWORKS Sustainability Associate, Dassault Systèmes (Nov 2024)
- 3DSwymer Associate, Dassault Systèmes (Dec 2024)
- Automotive Product Design, VirtualTester Online, Tangix Design & Development AB (2025 to 2026)
- Climate Change and Climate Finance Training, Bangladesh Institute of Management (BIM) / PKSF (Dec 2025)

## AWARDS & ACHIEVEMENTS

- Best Employee of the Quarter, Walton Hi-Tech Industries PLC. (Oct 2025), for the vacuum line modification and aging reduction project
- Best Trainer Award, Walton Hi-Tech Industries PLC. (Dec 2022), for welder skills training that cut joining leaks by 50%
- Merit Technical Government Scholarship, Rajshahi University of Engineering & Technology (2015)

## EXTRACURRICULAR ACTIVITIES

- Vice President, Barishal Divisional Association (2018 to 2019). Started a rental library for students in need.
- Youth Leader, Red Crescent Society, Oxford Mission High School. First aid provider and community volunteer.
- Active Citizen (Batch 43), British Council (2015). Group projects on local social issues.

## LINGUISTIC PROFICIENCY

- English: Listening C2, Reading C1, Spoken Production B2, Spoken Interaction B2, Writing B2
- Bengali: native language

## REFERENCES

**Dr. Md. Emdadul Hoque**
Professor and Thesis Supervisor, Dept. of Mechanical Engineering, RUET | mehoque@me.ruet.ac.bd | +880 1713228615

**Dr. Syed Mamun R. Rasid**
Professor, Dept. of Mechanical Engineering, RUET | mamun@me.ruet.ac.bd | +880 1330546246